

Technical Document #471: Cybersecurity - Comprehensive Overview

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Encryption converts plaintext into ciphertext to protect data confidentiality. AES and RSA are widely used encryption algorithms.

Penetration testing simulates cyber attacks to identify vulnerabilities. It helps organizations assess their security posture before real attacks occur.

Vulnerability management identifies, classifies, and remediates security vulnerabilities. Regular scanning and patching are essential components.

Multi-factor authentication (MFA) requires multiple forms of verification. It significantly reduces the risk of unauthorized access.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Key Takeaways:

- Penetration testing simulates cyber attacks to identify vulnerabilities.
- Vulnerability management identifies, classifies, and remediates security vulnerabilities.
- Encryption converts plaintext into ciphertext to protect data confidentiality.